INCH-POUND

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DEPARTMENT OF DEFENSE
STANDARD PRACTICE

IDENTIFICATION METHODS FOR BULK PETROLEUM PRODUCTS SYSTEMS INCLUDING HYDROCARBON MISSILE FUELS

AMSC N/A FSC 91GP

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. Comments, suggestions, or questions on this document should be addressed to AFPET/PTQ, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433 or e-mailed to AFPET.PTMT@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This standard provides a uniform method for the identification of liquid petroleum products and hydrocarbon missile fuels, rocket grade kerosene propellant, stored in and dispensed from Government-owned or operated storage facilities. The employment of this standard promotes greater safety to personnel and reduces the potential for error, confusion, and inaction in times of emergency by providing uniformity in the identification of products and product groups.

1.1.1 Mixing. The mixing of different grades and types of petroleum products and hydrocarbon missile fuels in field operations can be disastrous to the operation of aircraft, ships, ground vehicles, and support equipment. Improper identification, carelessness, and eradication of markings are often causes of such inadvertent mixing of different grades and/or dissimilar products.

1.1.2. Color assignments. MIL-STD-101, Color Code for Pipelines and Compressed Gas Cylinders, has assigned a color to each of six classes of materials. Color codes are intended to reduce the chances of accidental mixing of products during operation of permanently installed military bulk storage and dispensing systems. MIL-STD-101 permits only “yellow” as the identifying color for petroleum products and hydrocarbon missile fuels. All other colors will be removed or obliterated.

1.2 Applicability. The method of identification is applicable to all DoD activities. It is NOT applicable to aircraft and ships. Contractors at Government-leased facilities are encouraged to use this identification method.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include publications cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

NATO AFLP 1135	Interchangeability of Fuels, Lubricants, and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)
MIL-DTL-16884	Fuel, Naval Distillate
MIL-DTL-25524	Turbine Fuel, Aviation, Thermal Stable (JPTS)

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MIL-DTL-25576	Propellant, Rocket Grade Kerosene
MIL-DTL-43719	General Specification for Marking Materials and Markers, Adhesive, Elastomeric, Pigmented
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34) and NATO F-35

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-101	Color Code for Pipelines and for Compressed Gas Cylinders

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D910	Standard Specification for Leaded Aviation Gasolines
ASTM D975	Standard Specification for Diesel Fuel
ASTM D1655	Standard Specification for Aviation Turbine Fuels
ASTM D4814	Standard Specification for Automotive Spark-Ignition Engine Fuel
ASTM D4956	Standard Specification for Retroreflective Sheeting for Traffic Control

(Copies of these publications are available from https://www.astm.org.)

EUROPEAN COMMITTEE FOR STANDARDIZATION

EN 228	Automotive fuels. Unleaded petrol. Requirements and test methods
EN 590	Automotive fuels. Diesel. Requirements and test methods

(Copies of these publications are available from https://www.cencenelec.eu.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 704	Standard System for the Identification of the Hazards of Materials for Emergency Response

(Copies of this publication are available from https://www.nfpa.org.)

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SAE INTERNATIONAL

SAE AMS-STD-595	Colors Used in Government Procurement
SAE J1899	Lubricating Oil, Aircraft Piston Engine (Ashless Dispersant)

(Copies of these publications are available from https://www.sae.org.)

UNITED KINGDOM MINISTRY OF DEFENCE – (UK MOD)

DEF STAN 91-090	Gasoline, Aviation, Grades UL91, 100/130 and 100/130 Low Lead and 100/130 Very Low Lead and 100/130 Very Low Lead. JSD: AVGAS UL91, AVGAS 100, AVGAS 100LL and AVGAS 100VLL.
DEF STAN 91-091	Turbine Fuel, Kerosene Type, Jet A-1; NATO Code: F-35; Joint Service Designation: AVTUR

(Copies of these publications are available online at https://www.dstan.mod.uk.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Piping system. A piping system consists of any pipelines or part thereof used to convey liquid petroleum products including heating fuel and hydrocarbon missile fuels. This may be classified as tank car and tank truck loading and unloading connections; storage tank valves; pump manifold and valves; cross-country pipelines and their points of tie-in with pumping stations; oil tanker connections and manifolds and other similar dispensing outlets.

3.2 Storage system. All exposed fixed storage tanks not specifically exempted in 1.2.

3.3 Title. A title is the identification required on piping systems listed in 3.1. A title should identify the contents as identified in Table I, such as Gasoline, Aviation, Grade 100LL. Generally recognized abbreviations may be used in lieu of the title such as AVGAS 100LL for Gasoline, Aviation, 100LL. Refer to Table II for the appropriate NATO Code and DLA Product Code.

3.4 NATO code. A North Atlantic Treaty Organization (NATO) code number indicates that the product is interchangeable for its intended use with a particular product produced in one or more of the NATO participating nations. If there is no established NATO code, only product code and bands are required.

4. GENERAL REQUIREMENTS

4.1 Identification method. Bulk petroleum products and hydrocarbon missile fuels generally used in the military system have been classified and segregated by groups to facilitate the ready identification of product groups. This method establishes, defines, and assigns a yellow band or series of bands for recognition to each of eight groups of similar type products in a distinctive and conspicuous manner, as a visual aid and a supplement to written identification. These groups are aviation gasolines, automotive gasolines, jet fuels, distillates, heavy fuel (black) oils, lubricating oils, Aviation Thermally Stable Turbine

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Fuel, MIL-DTL-25524, and missile fuels as described in section 5 and Table I. The written identification consists of the exact title as defined in 3.3; the NATO symbol as prescribed and defined in 4.5; and the color band(s) as indicated in 5.1.3. The NATO code of a product (if established) is required on piping and storage systems located in the United States, as well as other NATO nations. NATO codes can be found in Table II and AFLP-1135.

4.1.1 Application of markings. Markings, (titles, bands, arrows and fire diamond (NFPA 704)), will be applied by painting and stenciling or, elastomeric film or reflectorized sheeting decals. If decals, film, or sheeting are used, material shall conform to specification MIL-DTL-43719 or ASTM D4956 as applicable. In addition to the locations indicated in the subsequent paragraphs, markings shall be applied at locations where the pipeline enters/exits the ground, in valve pits (refer to 5.1.8), at intersections of above ground piping, on above ground piping at approximately 60 meters (200 feet) intervals, all receiving connections, where line connections are made to manifolds, and at any other location necessary to assure ready identification of the product in the system. For single product, isolated systems, only the line into a manifold shall be marked. For multiproduct systems, product identification markings shall be placed on the inlet multiproduct line and all manifold outlet lines.

4.2 Color Specification. The colors assigned in the standard shall conform in hue and chroma to the requirements identified by numbers specified in SAE AMS-STD-595.

4.3 Employment of colors.

4.3.1 Warning color. The color Yellow No. 13655 is assigned as a primary warning for all flammable materials in accordance with the provisions of the basic color code, MIL-STD-101. Petroleum products and hydrocarbon missile fuels are considered falling within this classification of materials.

4.3.2 Use of black and white colors. The colors Black No. 17038 and White Gloss No. 17875 are assigned, WITHOUT SIGNIFICANT MEANING, for general use as indicated in this standard.

TABLE I. Product groups.

Product Group	Number of Yellow Bands	
	Wide	Narrow
Aviation Gasolines (AVGAS)	0	1
Automotive Gasolines	0	2
Turbine Fuels	0	3
Distillates	0	4
Heavy Fuel (Black) Oils	0	5
Lubricating Oils	Sign or Flag	
Aviation Thermally Stable Turbine Fuel (JPTS)	2	1
Missile Fuels	1	1

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TABLE II. Product specification with NATO and DLA product code.

Specification	NATO Code[1]	DLA Product Code[2]
ASTM D1655	F-24	JAA
MIL-DTL-83133	F-34	JP8
MIL-DTL-16884	F-76	F76
Def Stan 91-091	F-35	JA1
MIL-DTL-5624	F-44	JP5
MIL-DTL-25524	None	JTS
ASTM D910	F-18	100, 130
Def Stan 91-090	F-18	100
ASTM D975	None	DF1, DF2, DS1, DS2
EN 590	F-54	SFD
ASTM D4814	None	MRR, MUM, MUP, MUR
EN 228	F-67	ULG

NOTES:

1. Refer to AFLP 1135 for additional NATO Codes.
2. Contact DLA Energy Quality Technical Directorate at dlaenergyproducttechnology@dla.mil for DLA Product Codes.

4.3.3 Color limitations. UNDER NO CIRCUMSTANCES shall colors other than yellow for Warning and black and white for Identification be assigned to petroleum products and hydrocarbon missile fuels. The color RED is assigned exclusively for the use of fire protection materials and equipment. All other piping systems not carrying products within the scope of this standard will employ the warning colors assigned in MIL-STD-101 to the particular material.

4.3.3.1 Color of piping systems and storage systems. It is not the intent of this Standard to imply that the colors mentioned herein be used for coatings of piping or storage systems. The overall color shall be in accordance with departmental instructions.

4.4 Use of color bands. The use of color bands on all dangerous piping systems as established in MIL-STD-101 have been recommended as an aid to color-blind personnel since the use of bands will indicate that hazards are present. Identifying titles and bands on pipelines must be clearly legible at all times and should be frequently inspected to insure legibility through cleanliness of the marked area.

4.5 NATO code marking. Piping systems handling products for which a NATO code has been established, and are located in an area subject to servicing ground, sea, or air equipment of NATO countries will, in addition to the NATO code, include the appropriate U.S. Military Symbol (if established) as a part of the title in accordance with 5.1.1.

4.6 Facility Number. As a minimum, tanks will include the base assigned real property facility number. Follow the service specific real property identification documents for size and color.

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5. DETAILED REQUIREMENTS

5.1 Liquid petroleum and hydrocarbon missile fuels facilities system identification.

5.1.1 Titles. Exact identification on rigid piping systems and above ground fixed storage tanks is mandatory. If identification of tank trucks, semi-trailers or tank cars is desired or specified by departmental directives, the system described herein may be used.

5.1.1.1 Application of titles. Titles will be applied in such a manner as to be clearly visible from operating positions. The use of stencils with standard size markings specified in Table III is recommended. The black background will have a minimum border three-fourths inch wider than the lettered area. For piping smaller than three inches in diameter, metal flags or signs securely fastened to the pipe may be used with the appropriate title and products group band(s) lettered on the tag.

5.1.2 Color of NATO code identification. The appropriate NATO code and the broken line enclosure shall be of yellow color and placed in close proximity to the product title and bands (Figure 1).



FIGURE 1. NATO code marking.

5.1.3 Product group band(s). The yellow band(s) will be separated and distinguish the various groups of products. Except for lubricating oil lines (5.1.4) and multiproduct lines (5.1.5), petroleum products other than Aviation Thermally Stable Turbine Fuel (JPTS) and missile fuels are identified with one or more narrow bands. Aviation Thermally Stable Turbine Fuel (JPTS) is identified with a wide band, twice the width of a narrow band, followed by a narrow band, followed by another wide band. Hydrocarbon missile fuels are identified with one wide band, twice the width of the narrow band, followed by one narrow band. It must be re-emphasized that the title is the principal identifying feature and the band(s) is not to be relied upon to identify a particular product (Figure 2).

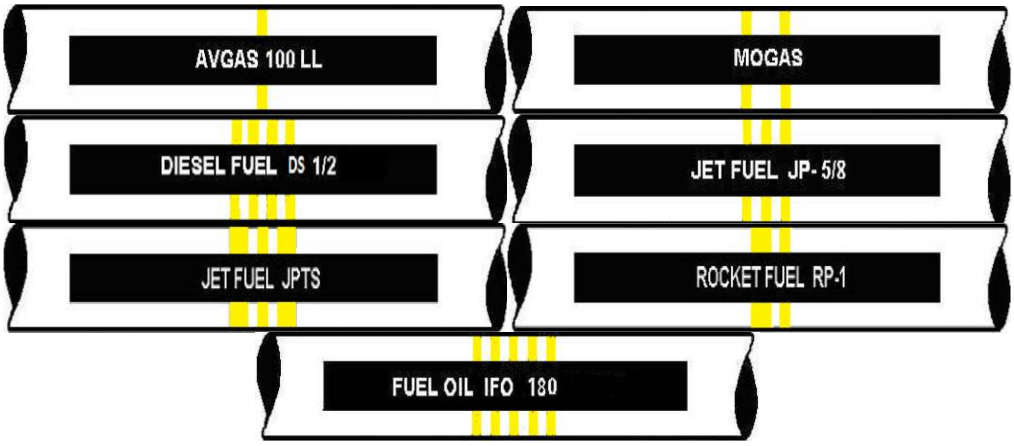


FIGURE 2. Product group band(s).

5.1.4 Lubricating oil lines. Because of the infrequency of its use, bands have not been assigned to lubricating oils. A flag or sign may be employed as illustrated in Figure 3. Each flag or sign shall have a yellow border of a minimum of three-fourths inch in width.

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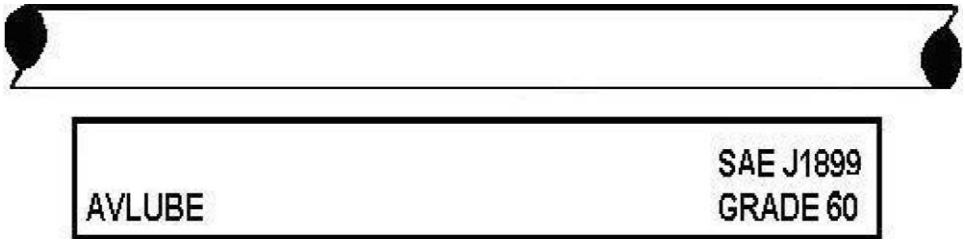


FIGURE 3. Lubricating oil lines.

5.1.5 Multiproduct lines. When a single pipeline is used for transporting more than one product, a flag or sign identifying the product currently in transit may be used in lieu of or as a supplement to the wide yellow identification band and nomenclature shown in Figure 4. Identification of the product will be made at the time of transfer. The yellow band in these instances will be a minimum of thirty-six inches in width.

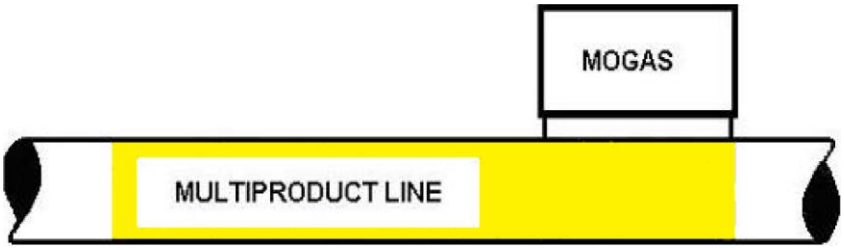


FIGURE 4. Multiproduct lines.

5.1.6 Use of arrows. An arrow painted in yellow may be used to indicate the direction of flow of the product in the line. It will appear adjacent to the title and band(s) as shown in Figure 5.

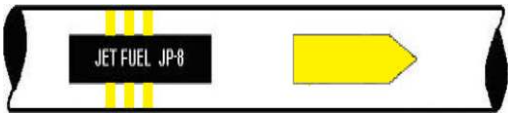


FIGURE 5. Use of arrows.

5.1.7 Application of flags. In instances where a piping system or tankage is buried or inaccessible, and only a valve stem and a wheel or gauging hatch are exposed, a metal flag or brass disc will be used as an aid identifying the product in the facility. The flag post may be permanently fixed to the pipeline or tank or in concrete adjacent to the structure. The brass identification disc will be placed on top of the valve wheel or gauging hatch.

5.1.8 Concrete valve and refueling pits. In concrete pits and similar conditions where space will not permit banding and stenciling of the pipe, the vertical band(s) will be painted on the wall adjacent to the pipe to represent a product group. The title of the product will be stenciled horizontally in white across the band(s). Where pit covers are installed, markings may be applied to the top of the covers.

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TABLE III. Sizes of letters and bands[1].

	Width of Bands		Space Between Bands	Length of Bands	Title Letter Size
	Wide	Narrow			
a. Pipe Diameter: Under 7.62 cm (3”)	15.2 cm (6”)	7.62 cm (3”)	7.62 cm (3”)	encircle	See Note 1
7.62 cm (3”) to 15.2 cm (6”)	15.2 cm (6”)	7.62 cm (3”)	7.62 cm (3”)	encircle	See Note 1
15.2 cm (6”) to 22.8 cm (9”)	15.2 cm (6”)	7.62 cm (3”)	7.62 cm (3”)	encircle	See Note 1
Over 22.8 cm (9”)	20.3 cm (8”)	10.1 cm (4”)	10.4 cm (4”)	encircle	See Note 1
b. Tank Capacity: Under 7,570 Liters (2,000 gal)	15.2 cm (6”)	7.62 cm (3”)	7.62 cm (3”)	83.3 cm (33”)	7.6 cm (3”)
Over 7,570 Liters (2,000 gal) to 10,000 bbls (420,000 gal) and under	15.2 cm (6”)	7.62 cm (3”)	7.62 cm (3”)	83.3 cm (33”)	15.2 cm (6”)
Over 10,000 bbls (420,000 gal)	20.3 cm (8”)	10.1 cm (4”)	7.62 cm (3”)	83.3 cm (33”)	30.4 cm (12”)

NOTES:

1. Pipeline letter size is provided in MIL-STD-101.

5.2 Special requirements for camouflaged systems. Bulk petroleum product piping and storage systems which have been camouflaged for concealment shall have the contained product title, grade, and NATO Code stenciled at the locations indicated in 3.1 and 4.1.1. Use the color Gray No. 36622 or Black No 37038, whichever is most easily discernable against the background color. Standard size letters as specified in Table III shall be used. Yellow color bands and markings which would detract from the camouflage effect shall not be utilized.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pipe identification marking system method called out in this standard is intended to promote safety and to reduce confusion as to pipe contents in the event of an emergency and to facilitate maintenance. It is also intended to increase the level of standardization among NATO nations for pipe and conduit identification in facilities used in ground petroleum products handling.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

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6.3 Exemptions from marking requirements. Department of Defense bulk petroleum product piping and storage systems can become targets for sabotage and other unlawful activities during times of civil strife. Facilities such as Defense Logistics Agency (DLA) fuel supply points and depots (Government Owned, Contractor Operated (GOCO's) and Contractor Owned, Contractor Operated (COCO's)) are particularly vulnerable because of their isolated locations and lack of security forces protection. The markings placed on large tanks (5.1.1 and Table III) in accordance with provisions of this standard can be readily seen from some distance and serve as an aid to persons contemplating destruction of government property. To preclude this possibility, exemptions from tank marking requirements will be approved upon written request of the cognizant agency or military department. The request/justification should be submitted to the cognizant agency or military department Service Control Point (Air Force Petroleum Office (AFPET), Navy Petroleum Office, or U.S. Army Petroleum Center (USAPC)).

6.4 Subject term (key word) listing.

Bands

Pipes

Product group bands

Warning colors

6.5 International standardization agreement implementation. This standard implements NATO AFLP 1135, *Interchangeability of Fuels, Lubricants, and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations*. When changes to, revision, or cancellation of this standard are proposed, the preparing activity must coordinate the action with the Lead Agent's Point of Contact for the international standardization agreement, as identified in ASSIST at https://assist.dla.mil.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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DLA – GS, PS

Preparing activity:
Air Force – 68
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